

Coordinate-wise SVRG for low signal-to-noise objectives

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One-page summary. Full details: [paper/technical_report.pdf](#); every claim and its limits: [docs/CLAIMS.md](#).

What it is. A variance-reduced optimizer with adaptive coordinate scaling, implemented in PyTorch and JAX, verified against an independent NumPy oracle, and studied on noisy finite-sum problems built to resemble tick data. The emphasis is on honest evidence: tuned baselines, held-out seeds, paired bootstrap intervals, and negative results reported as such.

Contributions.

1. **Three implementations that agree.** PyTorch, JAX and a NumPy oracle follow the same update rule, checked step for step by automated tests (including adversarial-input fuzzing); on a Colab T4 the PyTorch engine stayed within 1.8×10^{-7} of the oracle over 100 steps.
2. **Short proofs.** Unbiasedness and a variance bound for the control-variate gradient; a staleness bound valid for the clipped adaptive engine; a numerical-safety property; and a corollary of the Johnson–Zhang theorem that ties SVRG’s rate to the *preconditioned* condition number.
3. **A prediction from the theory, tested.** Jacobi-preconditioned SVRG with hyper-parameters read off the theorem and *no tuning* reached float64 round-off on 20/20 seeds of a badly scaled problem, needing $4.7\times$ fewer gradient evaluations than the project’s own adaptive engine, with measured per-epoch contraction 0.08 against the guaranteed 0.50.
4. **Fourteen experiments** against learning-rate-tuned Adam and SGD with momentum (constant and cosine-decay), plus ablations of the engine’s invariants.

Results at a glance (synthetic data unless stated).

Finding	Evidence
Variance reduction beats the best tuned baselines on well-scaled convex problems	20/20 (least squares), 12/12 (logistic) seeds
Condition number of badly scaled features after Jacobi scaling	$1.1 \times 10^7 \rightarrow 82$
Misaligned batches versus plain SGD (estimator error)	$2.1\times$ <i>worse</i>
Textbook Adam on adversarial float32 inputs (extreme but finite)	fails in 76% of trials; full guard set 0%
Snapshot-interval sensitivity	within $2.1\times$ of best for $16 \leq K \leq 256$

What did not work, stated plainly.

- **Real data:** on three small real datasets with raw features no method reached the target and Jacobi SVRG led the best tuned baseline by only 0.06–0.52 decades; after *standardising* the features, SVRG variants reached the target on 3/3 datasets and the baselines on 0/3, so the shortfall was raw-scale conditioning.
- **No wall-clock win** over a direct solve at $n = 40000$, $d = 200$ (normal equations 0.06 s, SVRG 0.08 s); **no benefit** for walk-forward signal tracking (IC differences at most 0.0054).
- **Non-convex network:** sharper stationarity but worse held-out loss; with validation-tuned rates the deficit tracks adaptive scaling, not variance reduction. The adaptive engine has **no convergence proof**; **no market data** was used; **GPU wall-clock is unmeasured**.

Corrections made along the way (logged in [CHANGELOG.md](#)): a claimed benefit of clamping over additive ε was retracted after an ablation; a “best on all datasets” statement was corrected to a tie; a missing convexity hypothesis was added to a theorem statement.

Verify it yourself.

- Tests: `pip install -r requirements.txt && python -m pytest`
- All experiments: `python -m experiments.run_experiments`
- Report numbers match stored results: `python paper/make_numbers.py --check`